AI + ML RESEARCH PROJECT GUIDE

A practical workflow for choosing a research problem, finding a genuine gap, running rigorous experiments, and producing a reproducible paper plus production implementation.

Personalized for Sai Ruthwik Reddy Kandimalla

Research question | Baselines | Experiments | Ablations | Reproducibility | Publication

Research formula: Real problem -> literature review -> gap -> research question -> hypothesis -> baseline reproduction -> controlled contribution -> experiments -> ablations -> robustness -> reproducible paper -> optional production deployment.

Research Project vs Production Project

The real-company use-case method is an excellent starting point for research, but a research project must produce new and defensible knowledge. Deploying a useful system demonstrates engineering; research must additionally show what was unknown, what was tested, and what conclusion is supported by evidence.

Starts with a business problem	Starts with a focused research question
Optimizes reliability, latency, cost and usability	Tests a hypothesis under controlled conditions
Original engineering can be sufficient	Requires a contribution or new empirical finding
Primary output is a deployed application	Primary output is evidence, paper, code and experimental artifacts
Testing focuses on system behavior	Evaluation includes baselines, robustness and ablation studies

A strong combined project

Research layer: investigate a measurable gap. Production layer: deploy the strongest method as a secure and observable AI/ML service.

Valid first research contributions

- A new method or modification
- A new benchmark, dataset or evaluation protocol
- A careful empirical comparison under neglected conditions
- A robustness or failure analysis
- A cost, latency or efficiency improvement
- A negative result that challenges an assumption
- Application of an existing method to a meaningful under-studied language or domain

Stage 1 - Choose a Narrow Research Area

Select an area related to the skills you are already building. Avoid broad titles such as "Improving Agentic AI." A useful topic names the system, condition, outcome and comparison.

Recommended areas for your background

- RAG retrieval, reranking and evaluation
- Agent reliability under tool and provider failures
- Long-term memory for stateful agents
- Human-in-the-loop decision policies
- Cost-aware model and tool routing
- Small-model fine-tuning and model compression
- Hallucination detection and grounded generation
- Multilingual and code-mixed Indian-language NLP
- LLM and agent security evaluation
- ML drift detection and automated retraining
- Multimodal document understanding

Topic narrowing template

How does [method or design choice] affect [measurable outcome] compared with [baseline] under [specific dataset, domain or failure condition]?

Examples

- How does memory summarization affect factual consistency and token cost in long-running support agents?
- Does uncertainty-based human escalation reduce unsafe tool actions compared with fixed-rule approval?
- Does hybrid retrieval with reranking improve faithfulness for Telugu-English technical support queries?
- Do graph-controlled workflows recover from tool failures more reliably than free-form ReAct agents?

Stage 2 - Conduct a Structured Literature Review

Company engineering blogs help reveal practical problems, but research decisions must also be grounded in scholarly literature.

Core research sources

Google Scholar	Broad paper discovery and citation tracing
Semantic Scholar	Related papers, influential citations and topic exploration
arXiv	Recent preprints in AI, ML and computer science
Papers with Code	Papers connected to code, datasets, tasks and benchmarks
ACL Anthology	NLP, LLM and computational linguistics research
IEEE Xplore / ACM Digital Library	Peer-reviewed engineering and computing literature

Define the review protocol before searching

- Research question and scope
- Publication years and accepted venues
- Inclusion and exclusion criteria
- Search keywords and databases
- Screening process
- Fields to extract from each paper
- Method for handling duplicate or conflicting evidence

Literature matrix columns

- Title, year and venue
- Research question
- Dataset and task
- Method and model
- Baselines
- Metrics and result
- Limitations
- Code and data availability
- Potential gap relevant to your project

Do not call “using a different framework” a research gap. The gap must concern capability, evidence, efficiency, robustness, fairness, safety, or generalization.

Stage 3 - Identify and Validate the Research Gap

A good gap is specific enough to test and important enough to matter. Validate it across several papers rather than relying on one author’s limitation section.

Strong gap patterns

- Performance is poor on multilingual, code-mixed or domain-specific data.
- Evaluation measures final answers but ignores tool trajectories or unsafe intermediate actions.
- A method improves quality but its latency or operating cost is impractical.
- Published results rely on clean benchmarks and omit noisy production conditions.
- The system fails when tools return stale, conflicting or incomplete information.
- The technique has not been tested under distribution shift or data drift.
- Memory methods increase context size without clear task-completion gains.
- Existing studies compare models but not recovery behavior under partial outages.

Gap validation test

Is the issue documented by multiple papers?	Yes, or supported by a convincing preliminary investigation.
Can it be measured?	There are clear outcomes, metrics and baselines.
Can you run it with available data and compute?	The scope fits your hardware, budget and timeline.
Would the answer matter?	It changes a design, method, evaluation or deployment decision.
Is your contribution distinct?	It is not merely changing libraries or UI.

Stage 4 - Form the Question and Hypothesis

The research question determines the entire experimental design. Keep one main question and, if needed, two or three supporting questions.

Question template

Does method X improve outcome Y compared with baseline Z under condition C?

Hypothesis template

Method X will improve metric Y because of mechanism M, but may worsen cost, latency or another trade-off.

Example

Question: Does hybrid retrieval plus cross-encoder reranking improve answer faithfulness compared with dense retrieval on code-mixed technical-support documents?

Hypothesis: Hybrid retrieval with reranking will improve recall and faithfulness but increase median response latency and cost.

Operationalize every term

Improves	Statistically and practically meaningful gain over baseline.
Faithfulness	A stated metric, rubric and human/automated judging protocol.
Latency	Median and tail latency measured under stated hardware and load.
Cost	Token or compute cost per successful task.
Failure	Explicitly defined tool, provider, retrieval or validation failure.

Stage 5 - Dataset, Baseline and Reproduction

Choose data because it answers the research question, not because it is convenient. Whenever possible, combine one established benchmark with one realistic dataset for generalization.

Dataset selection checklist

- Task and labels match the research question.
- License permits your intended use and release.
- Size is sufficient for meaningful comparison.
- Train, validation and test protocol is defensible.
- Biases, missing values and annotation quality are documented.
- The dataset version and provenance can be recorded.
- Sensitive or personally identifiable data is excluded or properly handled.

Baseline reproduction

- Implement a recognized baseline before proposing an improvement.
- Match the original split, preprocessing and metrics where possible.
- Record differences in model version, data, hardware and hyperparameters.
- Verify that your result is reasonably close to the reported baseline.
- Freeze the baseline before testing the proposed contribution.

Reproduction, replication and robustness

Reproduction	Use the same data and method to obtain a comparable result.
Replication	Apply the method to new data and seek a similar conclusion.
Robustness test	Change conditions, parameters or protocols to test conclusion stability.

Stage 6 - Design Controlled Experiments

Change one major component at a time. Otherwise, you will not know which change caused the result.

Experimental specification

- Independent variable: the method or component being changed.
- Dependent variables: quality, reliability, latency, cost or other outcomes.
- Controlled variables: model, prompt, data split, hardware and budget.
- Baselines: simple, strong and relevant comparisons.
- Random seeds and number of repeated runs.
- Hyperparameter selection method.
- Hardware and software versions.
- Statistical and practical significance criteria.

Agent-specific evaluation

- Task-completion rate
- Tool-selection accuracy
- Tool-argument correctness
- Unnecessary action count
- Recovery from tool failures
- Unsafe-action rate
- Human-escalation precision and recall
- Latency, token usage and cost per successful task

ML-specific evaluation

- Appropriate classification or regression metrics
- Calibration and threshold analysis
- Subgroup and fairness analysis where relevant
- Error taxonomy
- Inference latency and memory usage
- Performance under distribution shift

Always report more than the best average score. Include variance, failure cases, sample size, and the trade-off between quality, latency and cost.

Stage 7 - Ablations, Robustness and Analysis

Ablation studies identify which component caused the improvement. Robustness tests establish whether the conclusion survives realistic changes.

Example RAG ablation ladder

- Dense retrieval only
- Hybrid retrieval
- Hybrid retrieval plus reranking
- Add query rewriting
- Remove metadata filtering from full system
- Compare chunking strategies and sizes
- Swap the generator while freezing retrieval

Robustness conditions

- Noisy, incomplete or ambiguous input
- Out-of-domain examples
- Different datasets or languages
- Different model families
- Tool and provider failures
- Adversarial or injected instructions
- Long context and conversation history
- Class imbalance and distribution shift
- Reduced compute or token budget

Analysis checklist

- Does the result support the hypothesis?
- Is the improvement statistically and practically meaningful?
- Which component produced the gain?
- Where does the proposed method fail?
- Does it generalize to another dataset or model?
- What is the latency, cost and complexity penalty?
- Could leakage, judge bias or tuning choices explain the result?

A carefully measured negative result is still valid research. Do not hide it. Explain what assumption failed and what future work follows.

Stage 8 - Reproducible Research Engineering

Another researcher, or your future self, should be able to recreate every important table, figure and model result.

Track and version

- Source code and commit hash
- Dataset name, source, version and preprocessing
- Model, tokenizer and embedding versions
- Prompts, tool schemas and workflow configuration
- Hyperparameters and random seeds
- Dependency lockfile and container image
- Hardware and runtime configuration
- Experiment logs, metrics and artifacts
- Exact script that generated each table and figure

Recommended repository structure

data/README.md	Sources, versions, licenses and preparation instructions
src/	Reusable research implementation
configs/	Versioned experiment configuration
scripts/	Data preparation, training and evaluation commands
tests/	Unit and integration checks
results/	Tables, figures and machine-readable summaries
paper/	Manuscript and references
Dockerfile / lockfile	Reproducible environment

Reproducibility exit test

A fresh environment can reproduce the principal experiment and regenerate the key tables/figures by following one documented command sequence.

Stage 9 - Write the Paper

Organize the report so that the research question, evidence and contribution remain clear.

Abstract	Problem, method, main result and contribution in compact form.
Introduction	Motivation, gap, question and contributions.
Related work	Position the project against relevant literature.
Methodology	Describe baseline, proposed method and assumptions.
Dataset	Provenance, licensing, splits, bias and preparation.
Experimental setup	Models, prompts, hardware, parameters and evaluation protocol.
Results	Main comparison with uncertainty and significance.
Ablations and robustness	Explain causes and boundary conditions.
Discussion	Interpretation, trade-offs and implications.
Limitations and ethics	Weaknesses, risks, bias, privacy and misuse.
Conclusion	Answer the question without overstating evidence.
Reproducibility appendix	Commands, configurations and supplemental details.

Contribution statement

Write three concise bullets: what problem you investigate, what method/evaluation you contribute, and what the evidence demonstrates.

Best Combined Research + Production Project

For your career goal, the most valuable project combines scientific investigation with production engineering.

Recommended topic

Reliable memory and failure recovery for long-running enterprise support agents

Research layer

- Compare full-history, sliding-window, summarized, vector and importance-ranked memory.
- Inject tool failures, stale retrieval and provider timeouts.
- Measure task completion, factual consistency, recovery, latency, token use and cost.
- Run ablations to identify the contribution of memory and recovery components.
- Validate on a second support or technical-document dataset.

Production layer

- LangGraph with checkpointing and human approval
- FastAPI with async streaming and authentication
- PostgreSQL/pgvector plus Redis
- Docker and CI/CD
- AWS deployment
- Evaluation pipeline, tracing and alerts
- Prompt-injection controls and access-filtered retrieval
- Provider fallback and idempotent tool execution

Deliverables

- Research paper and literature matrix
- Reproducible experiment repository
- Evaluation dataset and metric definitions
- Deployed demonstration
- Architecture and threat-model diagrams
- Model/system card and limitation statement
- Short presentation and resume-ready results

Five Research Topics Tailored to You

Agent reliability under tool failures	ReAct vs graph workflow vs retries/reflection/human approval	Completion, recovery, unsafe calls, latency, cost
Cost-aware model routing	Always-large vs rules vs classifier vs LLM router	Quality, routing accuracy, latency, cost
Long-running agent memory	Full, windowed, summarized, vector and ranked memory	Consistency, completion, tokens, latency
Human escalation policies	Always, never, rules, uncertainty and risk classifier	Unsafe actions, escalation precision, effort
Code-mixed RAG	Monolingual vs multilingual embeddings, hybrid search and reranking	Recall, faithfulness, correctness, latency

Recommended order

- Start with a controlled empirical comparison rather than inventing a new architecture.
- Use one strong baseline and one simple contribution.
- Complete the experiments and paper before expanding the production stack.
- Deploy only the strongest validated configuration.

Best first choice: Agent reliability under tool failures. It aligns with LangGraph, evaluation, observability, guardrails and production reliability while remaining experimentally measurable.

Research Readiness Checklist

- The research question is narrow and measurable.
- The gap is supported by multiple papers or preliminary evidence.
- The contribution is more meaningful than changing a framework.
- Dataset licenses and provenance are documented.
- A recognized baseline is reproduced first.
- Only one major factor changes per core experiment.
- Metrics match the task and include uncertainty.
- Ablations explain which component matters.
- Robustness tests cover realistic failure conditions.
- Negative and contradictory results are reported.
- Code, data processing, prompts and configurations are versioned.
- Random seeds, model versions, hardware and dependencies are recorded.
- The main experiment is reproducible from a fresh environment.
- Limitations, ethics, bias and privacy are discussed.
- The conclusion does not exceed the evidence.

Final standard: A good research project does not only build something impressive. It asks a precise question, compares against credible baselines, measures the answer rigorously, and makes the evidence reproducible.

Reference resources

Papers with Code: research papers connected to tasks, code, datasets and benchmarks. https://paperswithcode.co/

Awesome Reproducible Research: curated reproducibility case studies, projects and resources. https://github.com/leipzig/awesome-reproducible-research

Kempner Institute Computing Handbook: reproducible code, environments, seeds, provenance and end-to-end workflows. https://handbook.eng.kempnerinstitute.harvard.edu/s2_swe_for_research/reproducible_research.html

PapersFlow literature review workflow: protocols, screening, extraction and transparent synthesis. https://papersflow.ai/blog/reproducible-literature-review-workflow

Resource details and availability should be rechecked when beginning the project.